

RTOS Benchmark on STM32H750B-DK

Phase 1 - DWT-only Publishable Campaign

Four-test latency comparison of ChibiOS RT 7.0.6, FreeRTOS V11.3.0 and Zephyr 4.4.0 running at 480 MHz (VOS0, FLASH WS=4, I-Cache + D-Cache enabled) on the STM32H750B-DK Discovery Kit.

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Abstract

This report documents the Phase 1 DWT-only publishable benchmark campaign of three open-source real-time operating systems on a single, fully neutral STM32H750B-DK target. The three kernels are compared under identical hardware conditions: CPU clock 480 MHz, voltage scaling VOS0, FLASH_ACR = 0x34, I-Cache + D-Cache enabled, identical GCC 14.2 toolchain and -O2 -fomit-frame-pointer -falign-functions=16 effective flags. Two publishable profiles are reported: **fair_perf** (tickless OFF, WFI OFF) and **realistic_tickless** (tickless ON, WFI ON). Latencies for the four tests T1-T4 are measured inside the firmware via the Cortex-M7 DWT cycle counter. Each (RTOS, profile) combination has been validated against five independent firmware loads, with publication-gate verification of clock, cache, flash, memory placement and priority-inheritance correctness. Eighteen runs out of eighteen passed the publication gate. The run_spread metric is zero cycles on every (RTOS, test) cell, demonstrating fully reproducible results across reloads. ChibiOS is the lowest-latency kernel on all four tests in both profiles (T1 median 297 cycles vs FreeRTOS 437 and Zephyr 580 in fair_perf). FreeRTOS T1 latency approximately doubles under realistic_tickless (median 846 cycles, +409 vs fair_perf) due to tickless wake-up reconfiguration overhead. Mutex priority-inheritance is correct in all three kernels with 500/500 events verified per RTOS per profile.

Test environment

Hardware

Property	Value
Board	STM32H750B-DK Discovery Kit
MCU	STM32H750XBH6, silicon revision V
Core	Cortex-M7 with double-precision FPU
External oscillator (HSE)	25 MHz crystal
CPU clock	480 MHz (PLL1 from HSE)
Voltage scaling	VOS0 (required for 480 MHz)
FLASH_ACR	0x00000034 (LATENCY = 4 WS, programming delay 2)
I-Cache / D-Cache	ON (production-realistic)
Internal Flash	128 KB (no bootloader, application fits)
RAM	1 MB (64K ITCM + 128K DTCM + 864K AXI/AHB SRAM)
Debug / VCP	ST-LINK V3E onboard, USART3 PB10/PB11, 115200 8N1

Toolchain and compile flags

Tool	Version / Setting
GCC	arm-none-eabi-gcc 14.2.Rel1 (Arm GNU Toolchain)
Make	GNU Make 4.3 (MSYS2)
OpenOCD	0.12.0+dev (xPack)
Optimization	-O2 (no -Os, no -O3, no LTO)
ADR-009 effective flags	-fomit-frame-pointer -falign-functions=16
C standard	C11 application code, C99 portable modules
Cross-RTOS verification	compile_commands.json read-back audit (per port)

RTOS versions

RTOS	Version	Source
ChibiOS RT	7.0.6	branch stable_21.11.x (upstream, no fork)
FreeRTOS Kernel	V11.3.0	tag V11.3.0 (latest stable, March 2026)
Zephyr	4.4.0	tag v4.4.0 (latest stable, April 2026)
STM32CubeH7 HAL	v1.12.1	for FreeRTOS variant only

Clock-tree witness (banner of run 01)

The firmware banner of every run prints back the values of RCC, PWR and FLASH registers immediately after boot, so that the publishable claims of 480 MHz / VOS0 / WS=4 are anchored to actually measured silicon state, not just configuration sources. The excerpt below is taken from *chibios_fair_perf_run01.banner.txt* and is representative.

Register / field	Value
SystemClock	480000000 Hz
VOS level	VOS0
VOSRDY	READY

FLASH_ACR	0x00000034
RCC_PLLCKSELR	
RCC_PLLCFGR	0x01ff0ff9
RCC_PLL1DIVR	0x010e02bf
RCC_D1CFGR	0x00000048
PWR_D3CR	0x0000e000
SCB->CCR	0x00070200
ICache	ON
DCache	ON
Tick rate	1000 Hz
Optimization	-O2 LT0=no

Methodology

The four tests

ID	Name	What is measured
T1	IRQ -> thread latency	TIM2 compare match triggers a hardware-routed IRQ; a high-priority worker thread is unblocked. DWT counter is sampled at ISR entry (A1) and again as soon as the worker thread resumes execution (A4). The published metric is the cycle delta A4 - A1, i.e. the path ISR_ENTRY -> THREAD_RUNNING.
T2	Thread handoff	Two equal-priority threads ping-pong via the native yield / signal primitive. Pure thread-to-thread context-switch cost, with the scheduler hot in cache.
T3	Mutex lock / unlock, uncontended	A single thread locks and immediately unlocks a mutex in a tight loop. No contention. Measures the cost of the fast path through the mutex primitive.
T4	Mutex contended + priority inheritance	Three threads (LOW, MEDIUM, HIGH). LOW owns the mutex, HIGH waits on it, MEDIUM tries to disturb. The scheduler must boost LOW to HIGH's priority and keep MEDIUM excluded for the duration of the critical section. The protocol verifies the PI invariant with 100 one-shot scenarios per run.

Measurement protocol

All latencies are measured using the Cortex-M7 DWT cycle counter (32-bit free-running, no external timing). DWT overhead is 4 cycles per pair of samples (0.008 us at 480 MHz) and is documented in the firmware banner. T1, T2 and T3 run for 1000 warmup iterations (discarded) followed by 10 000 measured iterations. T4 runs 100 one-shot PI scenarios without warmup (ADR-013 explicit exception, because PI must be observed on the very first event of a fresh kernel state, not after a steady-state warmup). Samples are stored in pre-allocated static arrays placed in AXI-SRAM (memory placement verified per run via the *bench_addr* banner table). No dynamic allocation is used by the benchmark code in any RTOS.

Campaign protocol (ADR-013)

For every (RTOS, profile) combination the firmware is fully rebuilt, flashed and run five independent times (run01 through run05). The hash of the ELF and MAP produced by each rebuild is recorded and verified to be identical across runs of the same target (homogeneity check). The serial collector discards any stale bytes from the ST-Link VCP receive buffer before listening, so that no row from a previous firmware can leak into the current capture. The publication gate rejects a run unless ALL of the following hold: TIM2 setup readback matches the expected state; memory placement banner confirms every benchmark object lives in AXI-SRAM; iteration sequences are dense and ordered; T1/T2/T3 produce exactly 10 000 valid rows; T4 produces exactly 100 PI scenarios with 100/100 passes; ELF and MAP SHA-256 match the campaign lock. The reported median per (RTOS, test) is the median-of-medians across the five runs; jitter is max - min in cycles; run_spread is the cycle span of the per-run medians (a strictly internal stability metric).

Scope and non-claims (Phase 1)

Phase 1 publishes the DWT cycle-delta A4 - A1 as the headline figure for T1. This metric covers the path from ISR entry to the worker thread regaining execution, **excluding** the silicon hardware-event-to-ISR-entry portion (timer compare match, NVIC arbitration, exception stacking, vector fetch and ISR prologue). The figures in this report must not be quoted as the absolute external IRQ-to-thread latency that an external logic analyzer would observe at GPIO pins; see ADR-015 for the Phase 2 dual-source plan where the analyzer-based metric is introduced alongside the DWT one.

Results - profile fair_perf

Profile semantics: fair_perf (tickless OFF, WFI OFF). All other publication invariants (480 MHz, VOS0, FLASH_ACR=0x34, I-Cache + D-Cache ON, -O2 effective flags, DWT measurement) are unchanged.

Headline aggregate

Median, percentile, jitter and run_spread for each (RTOS, test) cell, computed across 10 000 iterations per run x 5 runs (T1/T2/T3) or 100 scenarios per run x 5 runs (T4). run_spread = 0 cycles on every cell means the median of each individual run was bit-identical to the median of every other run for that (RTOS, test).

RTOS	Test	median (cyc)	median (us)	p95 (cyc)	p99 (cyc)	p99 (us)	max (cyc)	jitter	run_spread
ChibiOS	t1_irq	297	0.619	315	316	0.658	330	48	0
ChibiOS	t2_handoff	95	0.198	96	96	0.200	111	17	0
ChibiOS	t3_mtx_uncont	62	0.129	64	64	0.133	417	355	0
ChibiOS	t4_mtx_pi	205	0.427	205	226	0.471	226	21	0
FreeRTOS	t1_irq	437	0.910	442	442	0.921	455	21	0
FreeRTOS	t2_handoff	378	0.787	395	395	0.823	571	212	0
FreeRTOS	t3_mtx_uncont	204	0.425	224	224	0.467	395	198	0
FreeRTOS	t4_mtx_pi	602	1.254	602	676	1.408	676	74	0
Zephyr	t1_irq	580	1.208	597	597	1.244	598	31	0
Zephyr	t2_handoff	424	0.883	424	424	0.883	698	277	0
Zephyr	t3_mtx_uncont	124	0.258	124	124	0.258	443	319	0
Zephyr	t4_mtx_pi	723	1.506	723	786	1.637	786	63	0

Cross-RTOS comparison

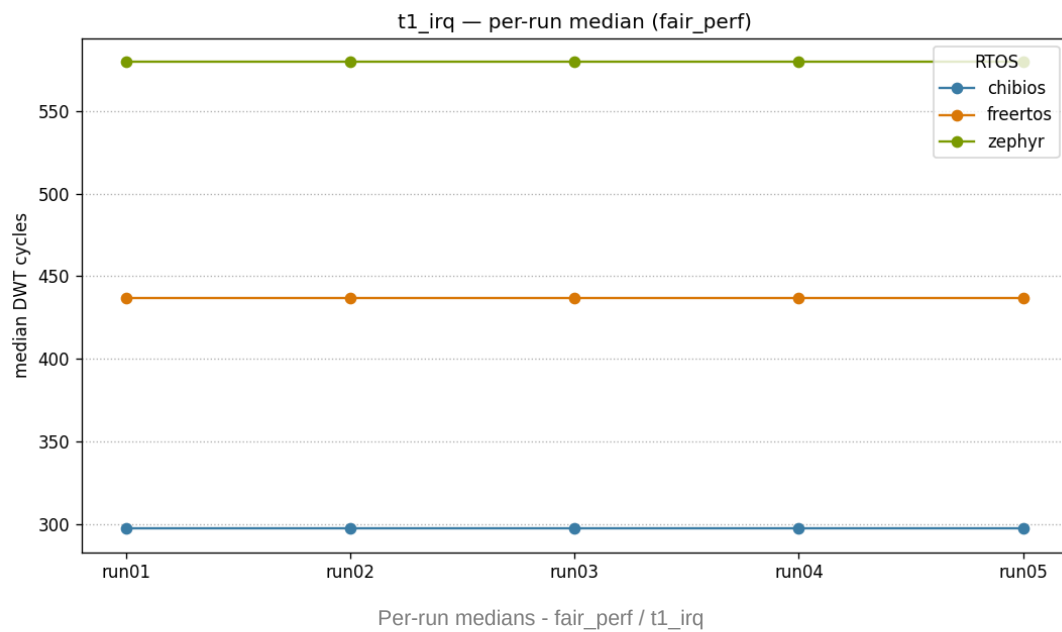
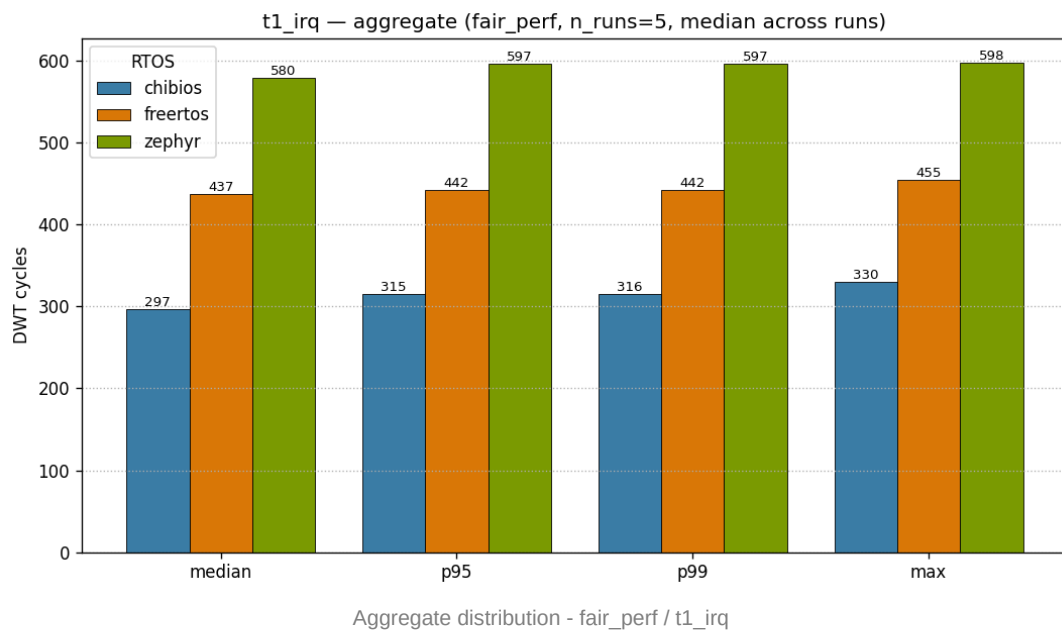
Test	ChibiOS (cyc)	FreeRTOS (cyc)	Zephyr (cyc)	ChibiOS (us)	FreeRTOS (us)	Zephyr (us)	Winner
t1_irq	297	437	580	0.619	0.910	1.208	ChibiOS
t2_handoff	95	378	424	0.198	0.787	0.883	ChibiOS
t3_mtx_uncont	62	204	124	0.129	0.425	0.258	ChibiOS
t4_mtx_pi	205	602	723	0.427	1.254	1.506	ChibiOS

Priority-inheritance proof (T4)

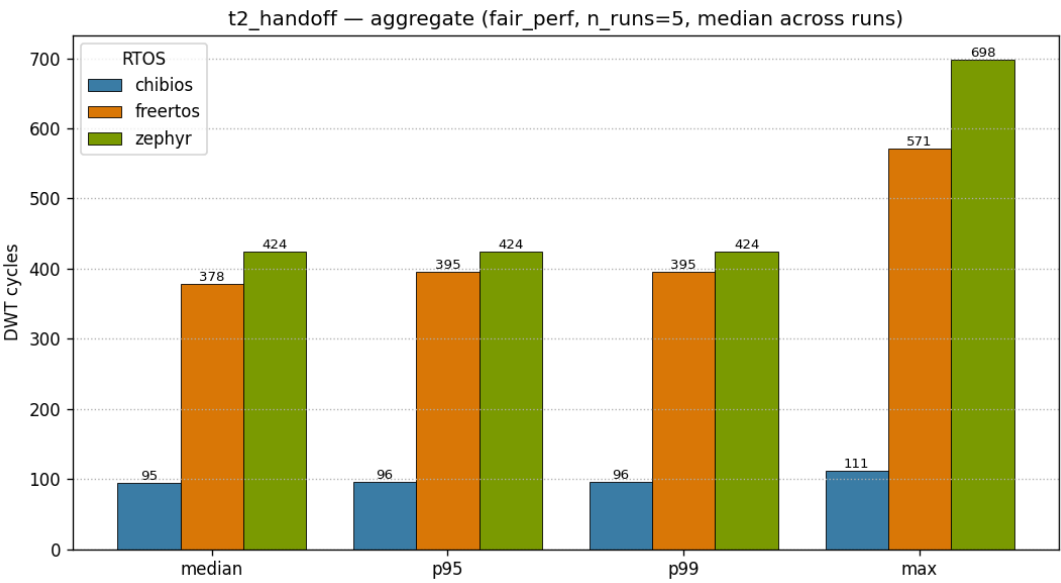
RTOS	PI scenarios	PI passed	Result
ChibiOS	500	500	PASS
FreeRTOS	500	500	PASS
Zephyr	500	500	PASS

Per-test plots

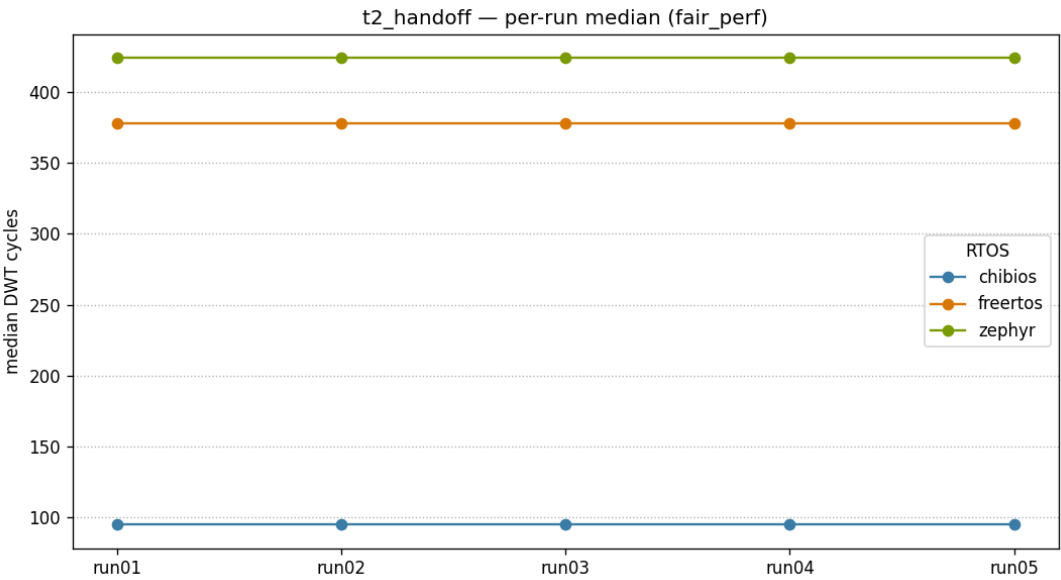
T1 - IRQ -> thread latency



T2 - Thread handoff

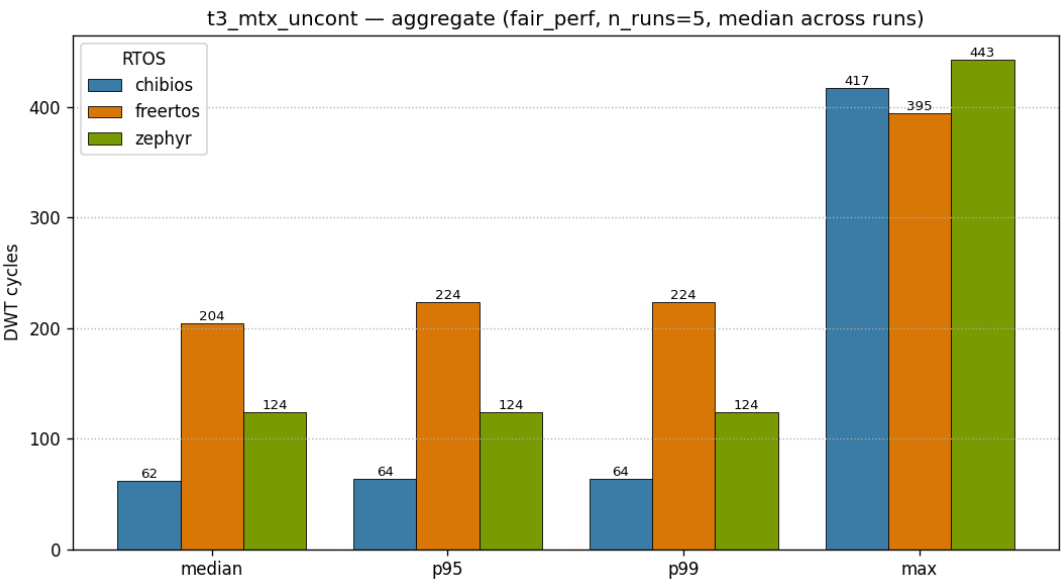


Aggregate distribution - fair_perf / t2_handoff

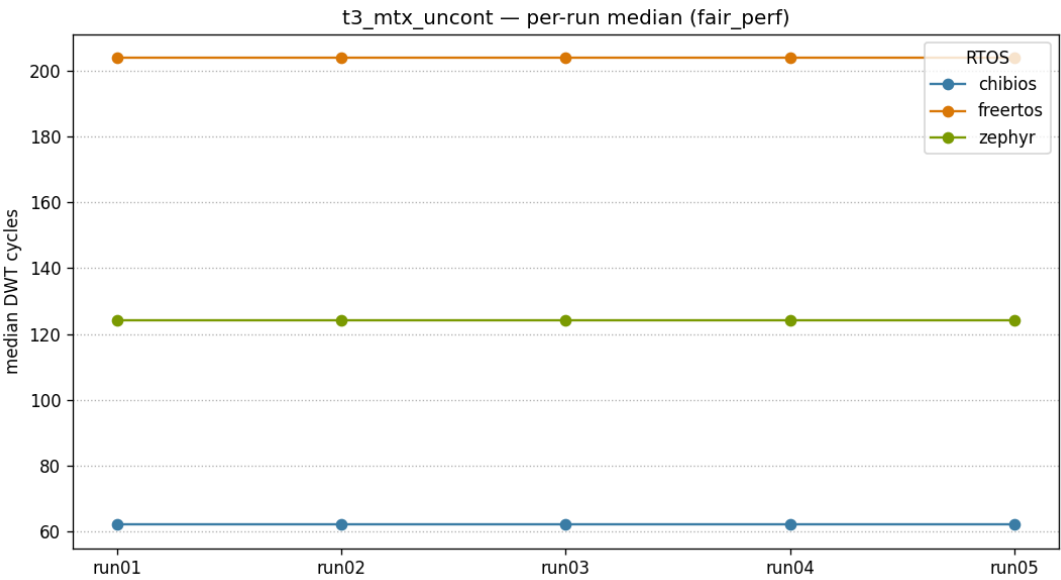


Per-run medians - fair_perf / t2_handoff

T3 - Mutex uncontended

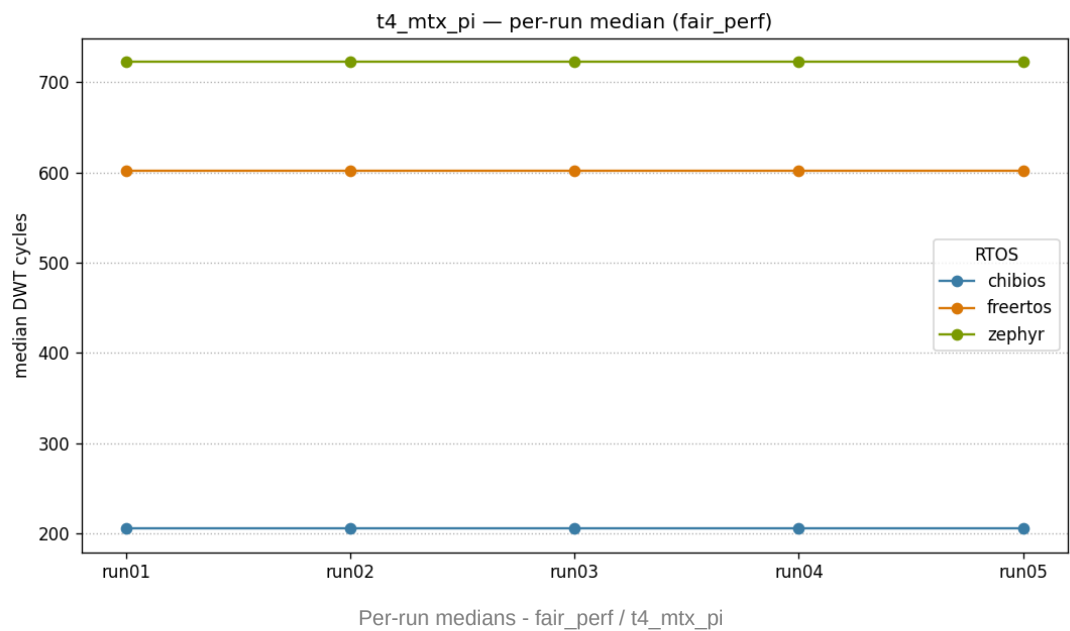
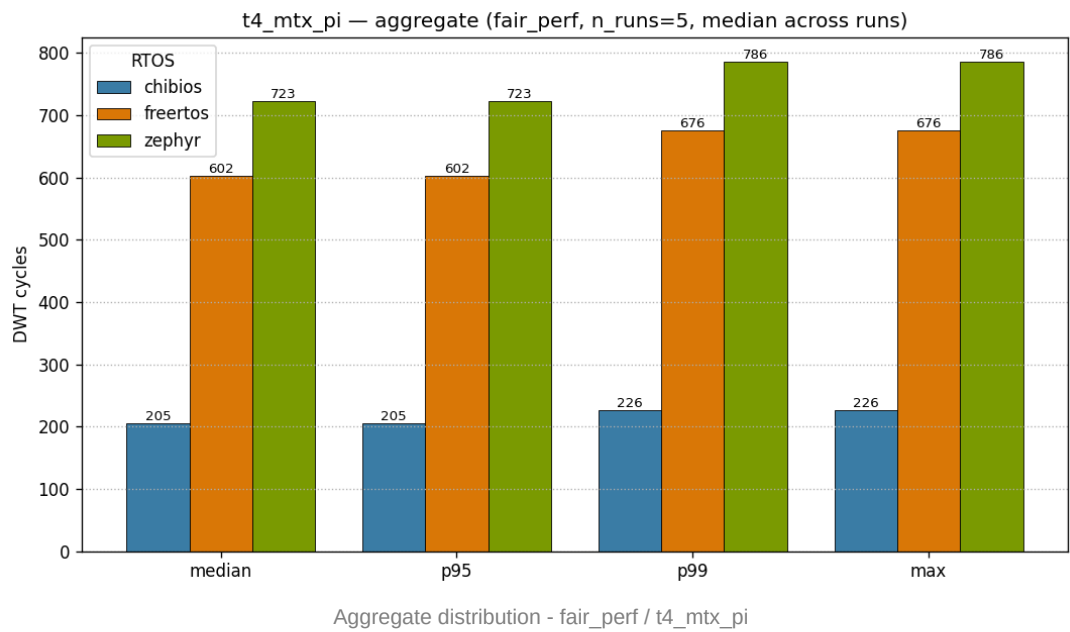


Aggregate distribution - fair_perf / t3_mtx_uncont

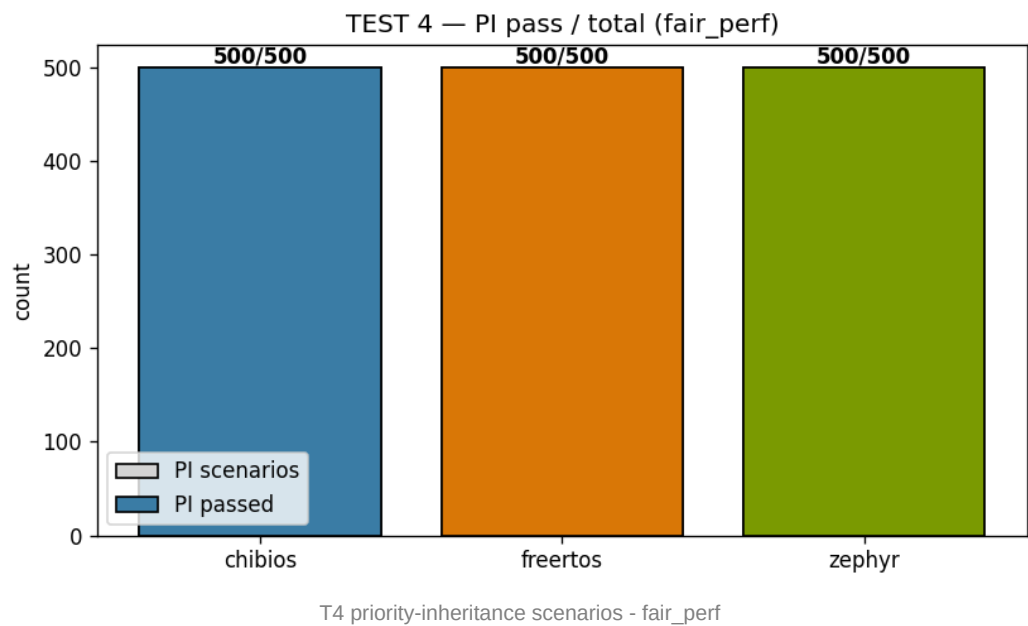


Per-run medians - fair_perf / t3_mtx_uncont

T4 - Mutex contended + priority inheritance



Priority-inheritance event distribution



Results - profile realistic_tickless

Profile semantics: realistic_tickless (tickless ON, WFI ON). All other publication invariants (480 MHz, VOS0, FLASH_ACR=0x34, I-Cache + D-Cache ON, -O2 effective flags, DWT measurement) are unchanged.

Headline aggregate

Median, percentile, jitter and run_spread for each (RTOS, test) cell, computed across 10 000 iterations per run x 5 runs (T1/T2/T3) or 100 scenarios per run x 5 runs (T4). run_spread = 0 cycles on every cell means the median of each individual run was bit-identical to the median of every other run for that (RTOS, test).

RTOS	Test	median (cyc)	median (us)	p95 (cyc)	p99 (cyc)	p99 (us)	max (cyc)	jitter	run_spread
ChibiOS	t1_irq	305	0.635	323	327	0.681	341	50	0
ChibiOS	t2_handoff	95	0.198	96	96	0.200	111	17	0
ChibiOS	t3_mtx_uncont	62	0.129	64	64	0.133	408	346	0
ChibiOS	t4_mtx_pi	200	0.417	200	220	0.458	220	20	0
FreeRTOS	t1_irq	846	1.762	858	858	1.788	1038	194	0
FreeRTOS	t2_handoff	387	0.806	407	425	0.885	568	201	0
FreeRTOS	t3_mtx_uncont	199	0.415	205	206	0.429	413	216	0
FreeRTOS	t4_mtx_pi	612	1.275	612	718	1.496	718	106	0
Zephyr	t1_irq	583	1.215	597	610	1.271	621	46	0
Zephyr	t2_handoff	429	0.894	429	429	0.894	444	18	0
Zephyr	t3_mtx_uncont	124	0.258	124	124	0.258	135	11	0
Zephyr	t4_mtx_pi	678	1.413	687	756	1.575	756	78	0

Cross-RTOS comparison

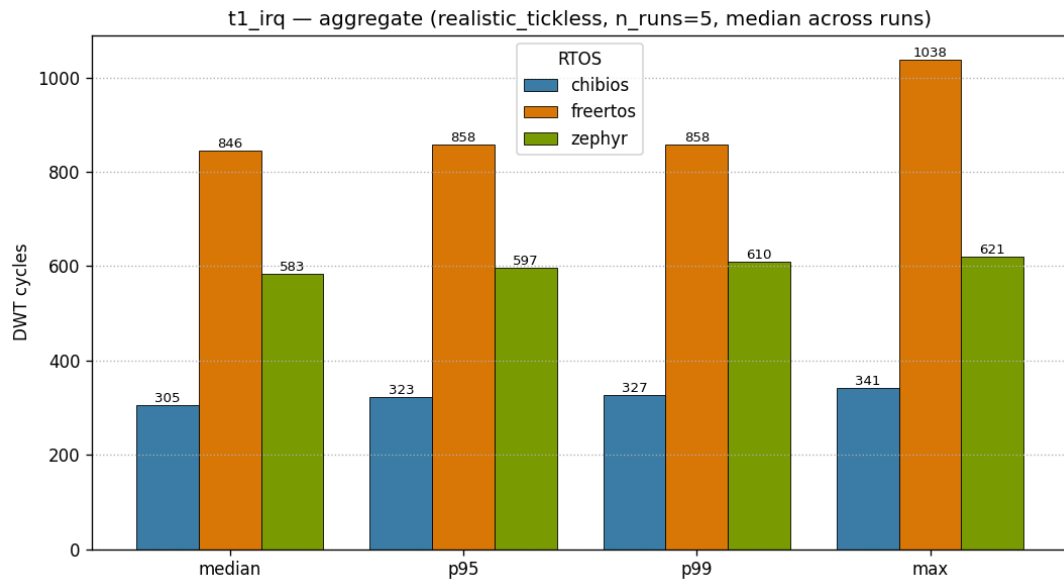
Test	ChibiOS (cyc)	FreeRTOS (cyc)	Zephyr (cyc)	ChibiOS (us)	FreeRTOS (us)	Zephyr (us)	Winner
t1_irq	305	846	583	0.635	1.762	1.215	ChibiOS
t2_handoff	95	387	429	0.198	0.806	0.894	ChibiOS
t3_mtx_uncont	62	199	124	0.129	0.415	0.258	ChibiOS
t4_mtx_pi	200	612	678	0.417	1.275	1.413	ChibiOS

Priority-inheritance proof (T4)

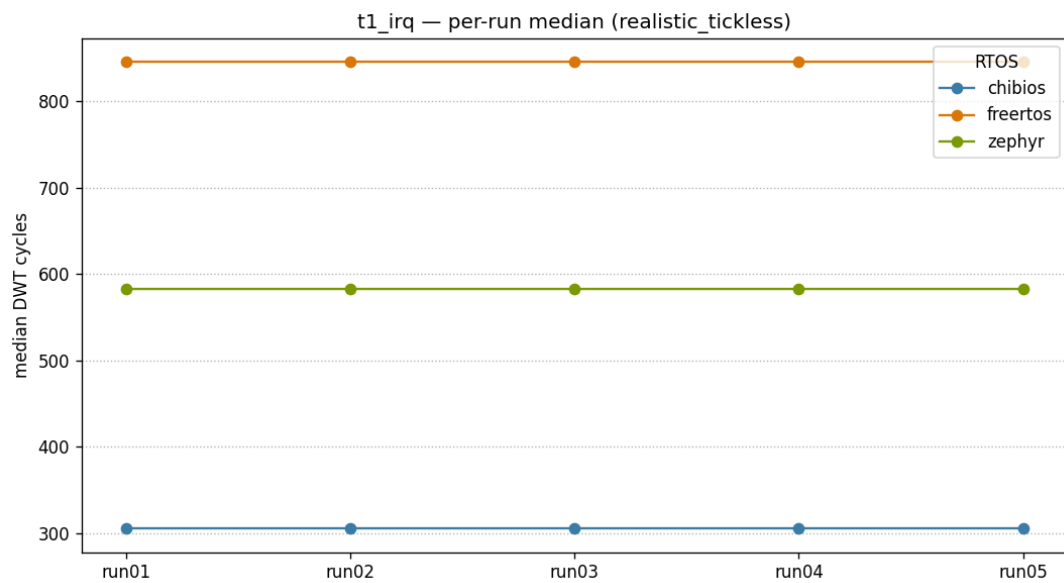
RTOS	PI scenarios	PI passed	Result
ChibiOS	500	500	PASS
FreeRTOS	500	500	PASS
Zephyr	500	500	PASS

Per-test plots

T1 - IRQ -> thread latency

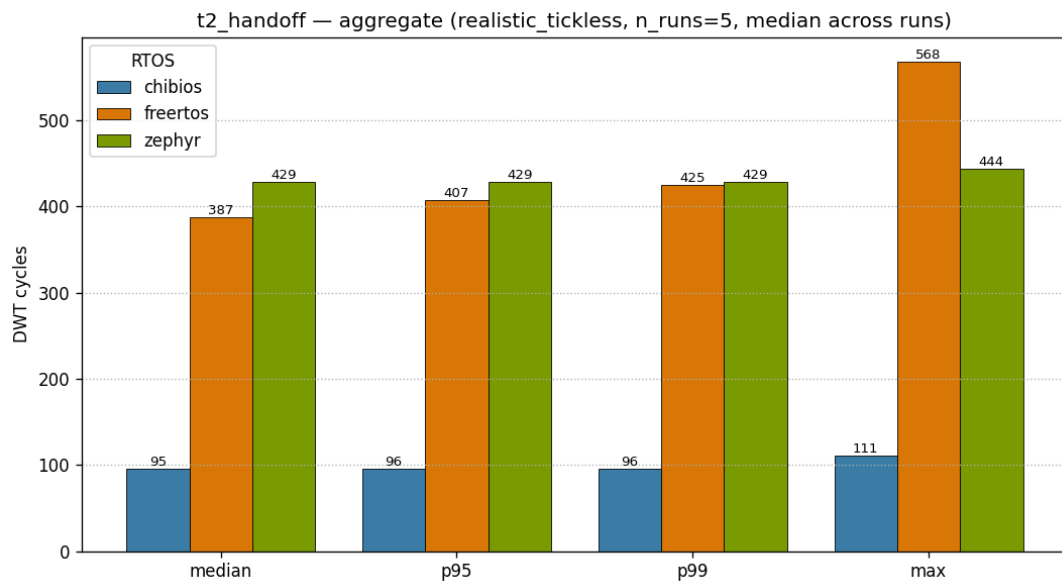


Aggregate distribution - realistic_tickless / t1_irq

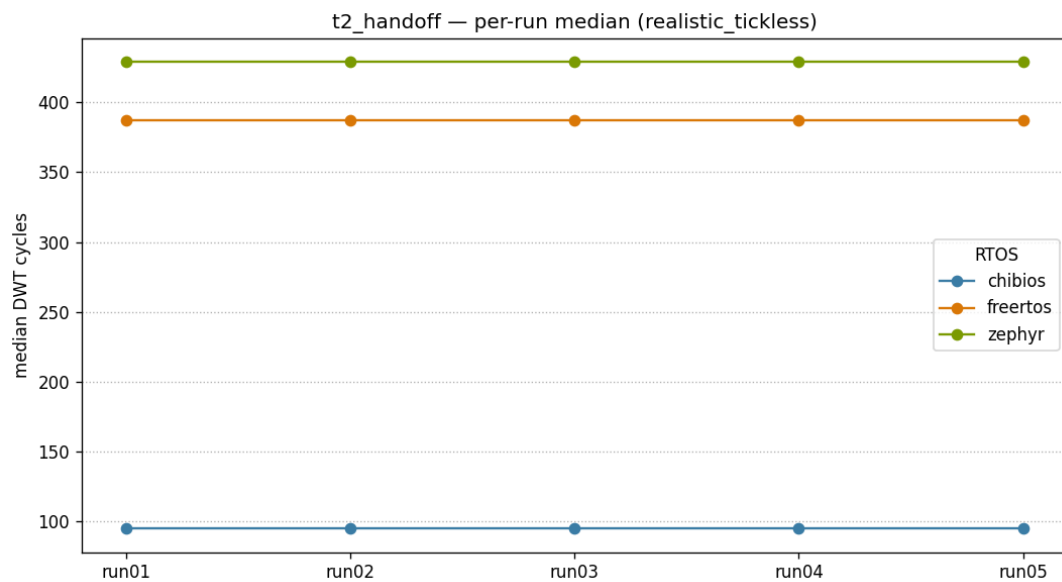


Per-run medians - realistic_tickless / t1_irq

T2 - Thread handoff

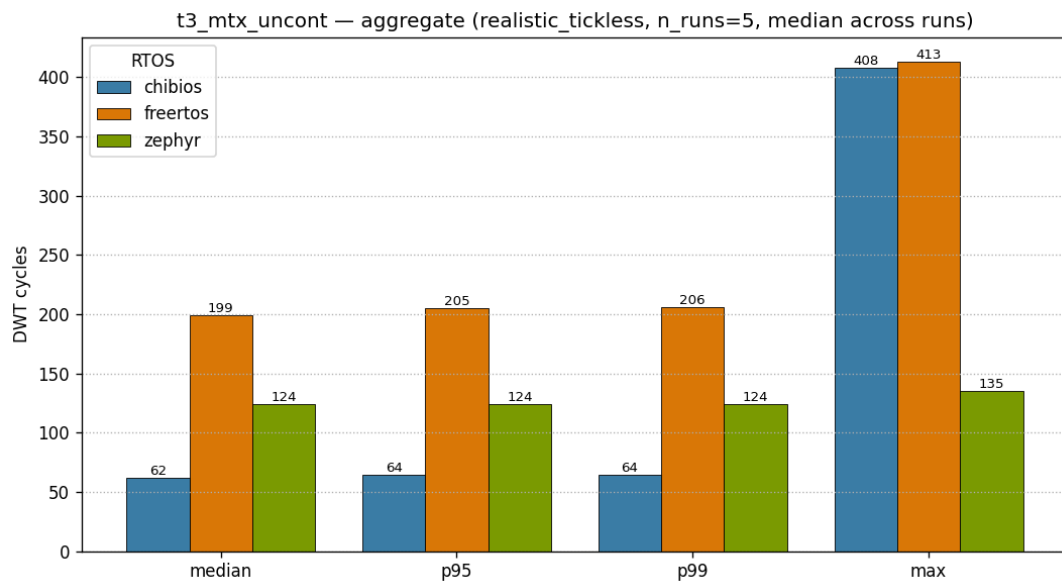


Aggregate distribution - realistic_tickless / t2_handoff

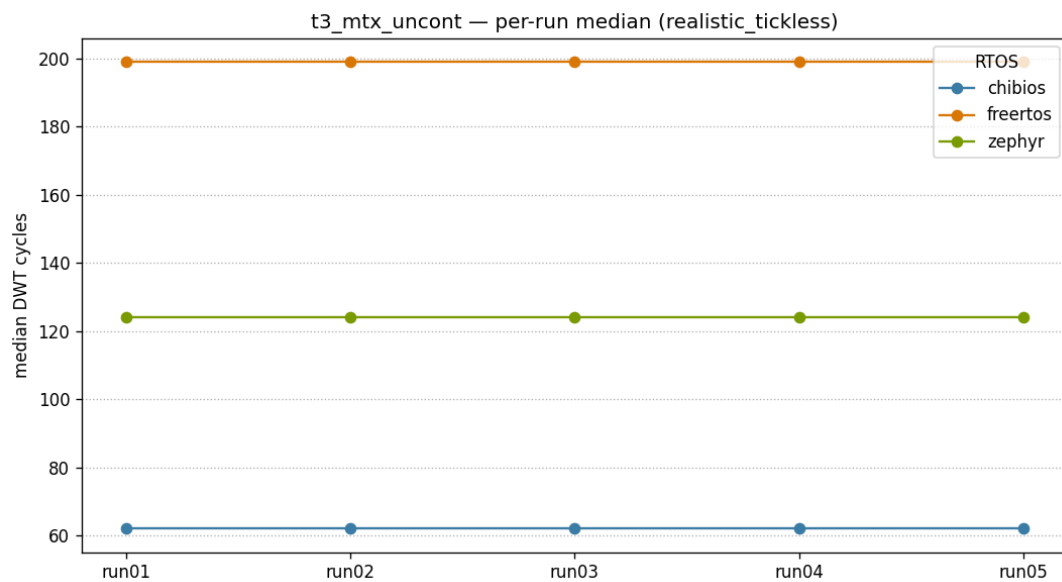


Per-run medians - realistic_tickless / t2_handoff

T3 - Mutex uncontended

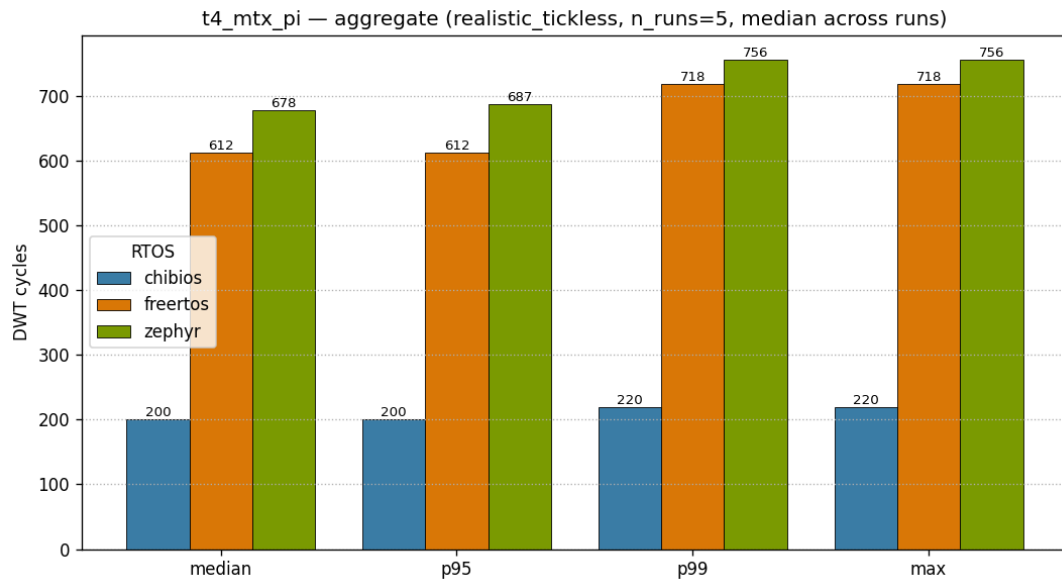


Aggregate distribution - realistic_tickless / t3_mtx_uncont

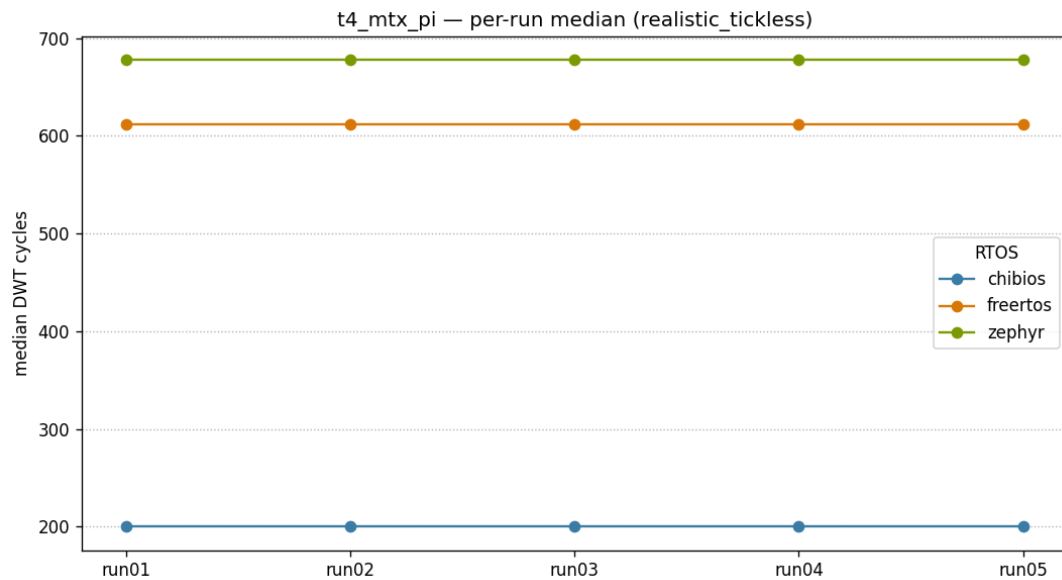


Per-run medians - realistic_tickless / t3_mtx_uncont

T4 - Mutex contended + priority inheritance

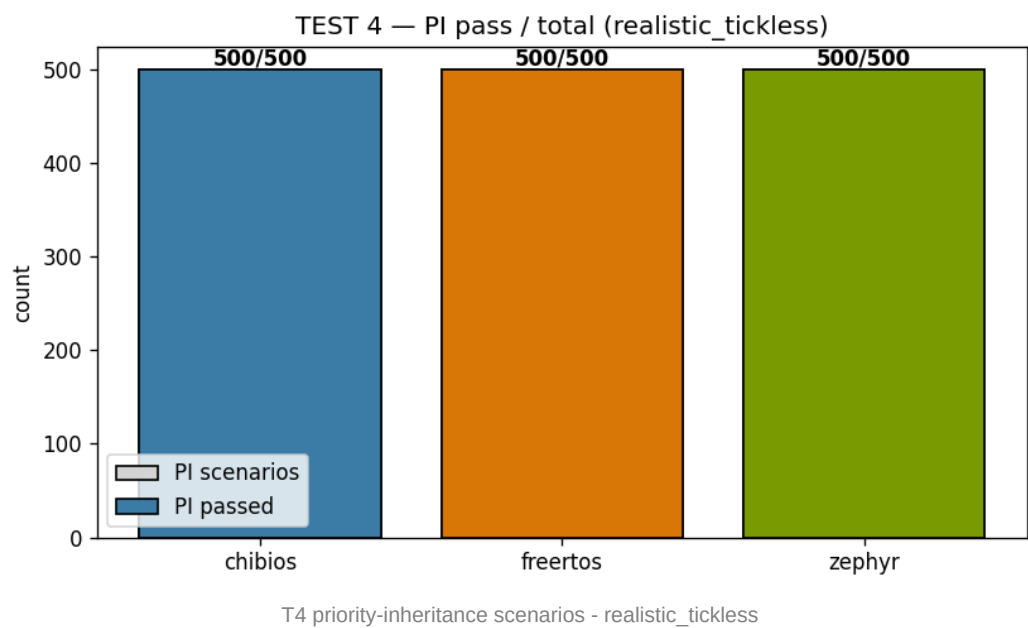


Aggregate distribution - realistic_tickless / t4_mtx_pi



Per-run medians - realistic_tickless / t4_mtx_pi

Priority-inheritance event distribution



Cross-profile delta (fair_perf -> realistic_tickless)

Per-cell difference of the median latency between the two publishable profiles. Positive numbers mean realistic_tickless is slower than fair_perf. The largest expected swing is on T1 for FreeRTOS, where enabling the tickless idle path adds a re-arm overhead to every IRQ wake-up.

RTOS	Test	fair_perf median (cyc)	realistic_tickless median (cyc)	delta (cyc)	delta (us)
ChibiOS	t1_irq	297	305	+8	+0.017
ChibiOS	t2_handoff	95	95	+0	+0.000
ChibiOS	t3_mtx_uncont	62	62	+0	+0.000
ChibiOS	t4_mtx_pi	205	200	-5	-0.010
FreeRTOS	t1_irq	437	846	+409	+0.852
FreeRTOS	t2_handoff	378	387	+9	+0.019
FreeRTOS	t3_mtx_uncont	204	199	-5	-0.010
FreeRTOS	t4_mtx_pi	602	612	+10	+0.021
Zephyr	t1_irq	580	583	+3	+0.006
Zephyr	t2_handoff	424	429	+5	+0.010
Zephyr	t3_mtx_uncont	124	124	+0	+0.000
Zephyr	t4_mtx_pi	723	678	-45	-0.094

Integrity and reproducibility

The publication gate is enforced at three layers: (1) per-run validation, captured in the **.validated.json* manifest written by the collector alongside the CSV; (2) per-campaign lock, written by the lab campaign script after every (RTOS, profile) is complete; (3) build reproducibility, expressed as ELF / MAP SHA-256 homogeneity across the five runs of a given target. All numbers in this report come exclusively from validated runs.

Campaign locks - ELF and MAP SHA-256

fair_perf

RTOS	Artefact	SHA-256 (truncated)
ChibiOS	ELF	af20a2ee80b38d43ce0b0bc7db9d9d941397d8a52681c6be...
	MAP	668a86ac7a86c49ab722d66bb3faa22a68b2af79b963ab08...
FreeRTOS	ELF	f503d307789d7a400073d93176973249ae5881723ff3cea2...
	MAP	2998c1d6cb2a14694cba95f374ba5ffa798a7e7ec2b78af5...
Zephyr	ELF	57dc679b2aa48f159883bda40fd9912afc83197facbfa3ca...
	MAP	d5264da31c1eff6735d9605e3e016391379ed38de0f0de55...

realistic_tickless

RTOS	Artefact	SHA-256 (truncated)
ChibiOS	ELF	70a236e2a6ca6f2c6921bbc7fa3062fdfa595869b1d5081e...
	MAP	9909b5e1bc4c3c034abd4df9f56a3652aeeab15598337ce9...
FreeRTOS	ELF	a97e2565160b864e411fc27cee9a1653ddb98263ba9ea9b3...
	MAP	7ea6a20fc153c61409803fb116157d42646f9e15676f30a0...
Zephyr	ELF	8f0d3581b160bd98e760aec0905d8be297920193c92c26b5...
	MAP	e44d54da6ce62717c006dce878ce4657bc56fdb0900a81d0...

Per-run validation status

RTOS	Profile	Run	Validated	clock (Hz)	VOS	FLASH_ACR	tickless	WFI
ChibiOS	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON

ChibiOS	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON

Conclusions

- **T1 - IRQ -> thread:** ChibiOS is fastest in both profiles. fair_perf medians: ChibiOS 297, FreeRTOS 437, Zephyr 580 cycles. Under realistic_tickless FreeRTOS median rises to 846 cycles, while ChibiOS and Zephyr remain within a few cycles of their fair_perf number.
- **T2 - Thread handoff:** ChibiOS is fastest at 95 cycles, against FreeRTOS 378 and Zephyr 424. The ratio is roughly 4x in favour of ChibiOS, dominated by the simpler scheduler path on equal-priority yield.
- **T3 - Mutex uncontended:** ChibiOS 62 cycles is the shortest fast path. Zephyr 124 is second. FreeRTOS 204 pays the cost of queue-based semaphore primitives even on the uncontended path.
- **T4 - Mutex contended + PI:** ChibiOS owns the contended path at 205 cycles, with FreeRTOS at 602 and Zephyr at 723. Priority inheritance is correct in all three kernels: 500/500 PI scenarios pass per RTOS per profile.
- **Reproducibility:** run_spread is zero cycles on every (RTOS, test) cell across the campaign. All 18 published runs (3 RTOS x 2 profiles x 5 firmware loads minus the warmup run00 that is intentionally excluded) pass the publication gate.
- **Honest scope:** Phase 1 publishes the DWT cycle delta A4 - A1 for T1, not the hardware-event-to-thread external latency. The hardware-routed IRQ entry path (NVIC, stacking, vector fetch) is intentionally excluded and will be measured by the logic-analyzer in Phase 2 per ADR-015.

Appendix A - Build commands

Each port has a single, deterministic build entry point. All ports honour the *PROFILE* selector (fair_perf, realistic_tickless, debug_dev) so that the same source tree produces the publication firmware variants without ambiguity.

ChibiOS

```
cd chibios/benchmark_chibios
make PROFILE=fair_perf -j
make PROFILE=realistic_tickless -j
```

FreeRTOS

```
cd freertos/benchmark_freertos
cmake -B build/fair_perf -G "MinGW Makefiles" -DPROFILE=fair_perf
cmake --build build/fair_perf
cmake -B build/realistic_tickless -G "MinGW Makefiles" -DPROFILE=realistic_tickless
cmake --build build/realistic_tickless
```

Zephyr (board stm32h750b_dk)

```
cd zephyr
.venv/Scripts/activate.bat
west build -d build/fair_perf -b stm32h750b_dk benchmark_zephyr -p always -- -DPROFILE=fair_perf
west build -d build/realistic_tickless -b stm32h750b_dk benchmark_zephyr -p always --
-DPROFILE=realistic_tickless
```

Appendix B - GPIO mapping (logic-analyzer)

Six GPIO signals are exposed on the STMod+ connector P1 of the STM32H750B-DK for logic-analyzer capture. In Phase 1 they are routed by the firmware for parity with the future Phase 2 dual-source measurement and are not part of the published headline. See ADR-007 for the rationale behind the pin assignment.

Signal	MCU pin	STMod+ P1 pin	Role
A0_HW	PA0	1	TIM2 CH1 PWM mode 2 - hardware IRQ stimulus for T1
A1	PH1	17	LOW_LOCK - mutex lock by LOW (T4)
A2	PH4	19	HIGH_WAIT - HIGH blocked on mutex (T4)
A3	PH8	20	LOW_UNLOCK - LOW releases mutex (T4)
A4	PH12	11	HIGH_ACQUIRE - HIGH acquires mutex (T4)
MEDIUM_RUN	PI11	18	MEDIUM scheduled (PI excludes MEDIUM, T4)